Fang Xu

Biomedical Engineering · ShanghaiTech University

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EDUCATION

ShanghaiTech University Sep 2023 - Expected Jul 2027

B.Eng. in Biomedical Engineering

Selected coursework: Biomedical Signals and Systems I; Biomedical Electronics I/II; Brain Science and Brain Disorders; Fundamentals of Micro/Nano Device Fabrication; Brain Organoids.

PUBLICATIONS

Fang Xu, Xingming Wen, Ze Xiong. **ARISE: Automated Routing and Integrated Simulation of Electrodes for Organoid Interfaces.** *Manuscript under review.*

Yuehan Wang, Haozhe Cui, Jiaying Du, Haichuan Di, Le Yan, **Fang Xu**, Kaiqi Cui, Yiduo Liu, Dongdong Tu. **Single-image 3D reconstruction of painted potteries using AI diffusion and feedforward models.** *npj Heritage Science 13, 545 · 2025.*

doi:10.1038/s40494-025-02114-x

RESEARCH EXPERIENCE

Wireless and Smart Bio-electronics (WiSe) Lab Sep 2025 - Present

Undergraduate Researcher · ShanghaiTech University

Advisor: Prof. Ze Xiong

ARISE · Organoid electrode-interface design

- Independently developed a pipeline from GDS scaffold geometries to electrode placement, graph-based routing, constrained interconnect selection, and four-sided pad layouts.
- Integrated batch geometry checks, coverage analysis, and COMSOL evaluation of contact pressure, strain, and material fraction; fabricated flexible electrode prototypes with Xingming Wen.

DREAM · Cross-subject EEG emotion recognition

Gold Prize · 11th National Undergraduate Biomedical Engineering Innovation and Design Competition

- Independently developed and optimized models for 30-channel, 10-second EEG segments, combining EEGNet-based spatiotemporal modeling, statistical features, MODMA-derived priors, and SEED-based stimulus prototypes.
- Designed functional-connectivity residual fusion, constrained within-subject ranking, and targeted boundary re-ranking; worked with the team on the final inference pipeline.

Park Lab Sep 2024 - Sep 2025

Undergraduate Researcher · ShanghaiTech University

Advisor: Prof. Sung Il Park

Adaptive wireless power transfer for optogenetics

- Developed an antenna-control strategy using RSSI and animal-posture features to predict optical power at an implanted receiver and select transmitting antennas in real time.
- Contributed to antenna impedance matching and antenna-array scanning and switching logic to address unstable power delivery.

Flexible wireless optoelectronic devices

- Independently completed the single-channel device workflow on flexible Cu/polyimide substrates, including photolithography, wet etching, micro-LED soldering, testing, and PDMS/parylene encapsulation.
- Collaborated with lab colleagues on the design, fabrication, soldering, testing, and encapsulation of four-channel devices.

SELECTED ACADEMIC PROJECTS

Ultrasound image analysis: classical methods to foundation models Mar - Jun 2026

Course project · Instructor: Prof. Xuejun Qian

- Compared four segmentation strategies and three classifiers on 120 breast ultrasound images with 36 radiomic features using stratified five-fold cross-validation. BUSAT+SVM achieved 89.2% accuracy; BUSAT+random forest achieved an AUC of 0.940.
- Benchmarked SAM2/MedSAM prompted segmentation and frozen foundation-model encoders for classification; BiomedCLIP+SVM achieved an AUC of 0.935. Also analyzed eight neck ultrasound acquisitions for carotid-artery quantification.

Single-image 3D reconstruction of painted pottery Jan - Oct 2025

Research project · Advisor: Prof. Dongdong Tu

- Contributed to data collection and experiments comparing diffusion-based novel-view synthesis and feedforward 3D reconstruction methods.
- Benchmarked seven reconstruction methods and evaluated appearance and geometry using PSNR, SSIM, LPIPS, Chamfer distance, and IoU.

Single-slope ADC for EEG acquisition May - Jun 2025

Course project · Instructor: Prof. Xiaochun Lai

- Designed and simulated a single-slope ADC module for an EEG acquisition front end in Cadence Virtuoso.
- Designed a folded-cascode OTA, sample-and-hold circuit, non-overlap clock, and linear ramp generator, focusing on circuit precision and timing.

ECG acquisition and real-time waveform display Dec 2024 - Jan 2025

Course project · Instructor: Dan Li

- Designed the PCB and coordinated fabrication for an ECG front end with an instrumentation amplifier, two gain stages, and a 50 Hz notch filter.
- Programmed an Arduino UNO for sampling and PC communication, integrating the analog front end with real-time waveform display.

INTERNSHIP EXPERIENCE

Shanghai Shangchuang Xinwei Investment Management Co., Ltd. Jul - Sep 2026

Investment Assistant · Shanghai, China

Conducted industry and technology research on semiconductors and embodied intelligence, examining value chains, product roadmaps, competitive positioning, and commercialization.

TECHNICAL SKILLS

Programming: Python, MATLAB, C/C++, PyTorch

Modeling & simulation: COMSOL Multiphysics, Abaqus, CST, AutoCAD, Solidworks

Hardware & design: Altium Designer, Cadence Virtuoso, L-Edit, Silvaco TCAD, SEMulator3D, Multisim, Keil µVision

Microfabrication: Photolithography, Etching, Micro-soldering, Encapsulation

Languages: Chinese (native), English (proficient), French (elementary)

TEACHING & SERVICE

Teaching Assistant 2025 - 2026

ShanghaiTech University

Teaching-assistant appointments in world archaeology (Fall 2026), management (Spring 2026), and non-fiction writing (Fall 2025).

Head, New Media Department Oct 2023 - Aug 2024

ShanghaiTech University Student Union